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**COMPARISON OF PRESSURE SWING
METHOD VS ENTRAINER METHOD
FOR PRODUCTION OF ANHYDROUS
ETHANOL USING ASPEN HYSYS**

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Abstract

This study explores the azeotropic distillation of ethanol-water azeotropic mixture using two separation techniques: Pressure Swing Distillation (PSD) and Extractive Distillation (ED) using cyclohexane as an entrainer. Given the limitations of conventional distillation in separating azeotropes, the two distillation methods were simulated with Aspen HYSYS to evaluate their economic performances.

The focus of the analyses was on total cost that include capital cost, operating costs, utility cost, and additional materials. From the simulation results, despite both methods having the same operating condition, the PSD method was shown to have lower total cost associated with raw materials, operating, and utility costs than ED method. The greater cost was a result of the additional solvent (cyclohexane) and the recovery processes associated with it, although initial capital cost for the ED method was similar to the PSD method.

This economic analysis indicates that under the operating conditions of this simulation, Pressure Swing Distillation can produce more economically viable separations for ethanol-water separation, making it favorable for applications where cost minimization is a priority.

Table of content

Abstract	2
Table of content	3
1.INTRODUCTION	5
1.1 Introduction:.....	6
1.2 Azeotropic distillation:.....	6
1.3 Ethanol:	7
1.4 Objectives:	8
2.LITERATURE REVIEW	9
2.1 Distillation process:	10
2.2 Azeotropic mixture:	10
2.3 Types of azeotropes:	12
2.3.1 Based on Boiling Behaviour:	12
2.3.2 Based on Phase Behaviour:.....	13
2.4 Azeotropic mixtures separation methods:.....	15
3. PROCESS AND METHODS	17
3.1 PRESSURE SWING METHOD:	18
3.2 Entrainer Method:	19
3.3 Cyclohexane:.....	20
4. SIMULATION AND SOFTWARE WORK	21
4.1 PROCESS SIMULATION:	22
4.2 ASPEN HYSYS:.....	22
4.3 PRESSURE SWING METHOD USING ASPEN HYSYS:	22
4.4 ENTRAINER METHOD USING ASPEN HYSYS:	24
5. ECONOMIC ANALYSIS AND RESULTS	26
5.1 Results:.....	27
5.2 Economic Analysis:	27
1-Cost Analysis:.....	27

2-Energy Consumption and Environmental Impact:	28
3-Product Quality and Production Rate:	28
6. DISCUSSION AND CONCLUSION	30
6.1 Discussion and Conclusion	31
References:.....	32
List of Figures:	33
List of Tables:	33

1.INTRODUCTION

1.1 Introduction:

In the azeotropic distillation process of the ethanol-water mixture, the formation of an azeotrope at certain compositions makes it difficult to achieve high-purity separation of the components. Aspen HYSYS, a process simulation software, provides a powerful platform for studying and optimizing such complex distillation processes. This project aims to explore the azeotropic distillation of the ethanol-water mixture using Aspen HYSYS to achieve high-purity ethanol separation.

In this introduction, we will provide an overview of the azeotropic nature of the ethanol-water mixture and the challenges associated with its separation. We will also highlight the importance of using advanced process simulation tools such as Aspen HYSYS to develop efficient distillation processes. Additionally, the introduction will outline the study's objectives and scope, emphasizing the significance of optimizing the distillation process to obtain high-purity ethanol.

By leveraging Aspen HYSYS, this project aims to investigate various process parameters, such as reflux ratio, feed stage location, and column configuration, to enhance the separation efficiency of the ethanol-water mixture. It will also provide an economic comparison between azeotropic distillation and the entrainer method

1.2 Azeotropic distillation:

Azeotropic distillation is a method for separating liquid mixtures and is widely used in industries such as chemical industry and oil refining. When the two components to be separated are azeotropic solution system or their volatility is very close, it is difficult to achieve the separation purpose by ordinary distillation method or the number of theoretical plates required is very large, and the reflux ratio is also large, which makes the equipment cost and operation cost too high and uneconomical. At this time, azeotropic distillation can be used. Azeotropic distillation is the addition of a third component to azeotropic solution, which can form azeotrope with one or more components in the original solution, and the volatility of the new azeotrope is significantly higher or lower than that of the original components. The content of each component in the new azeotrope is

different from that of the raw material solution, and the common concentrate can be used. According to whether the entrainer can form two liquid phases with one or several components in the system to be separated, azeotropic distillation can be divided into homogeneous azeotropic distillation and heterogeneous azeotropic distillation. The third component (azeotropic agent) is added to the heterogeneous azeotropic distillation to form two liquid phases with the original components, which are layered by a phase separator to azeotrope and target components. The separation is carried out and the recovered azeotrope can be recycled.

Homogeneous azeotropic distillation is commonly used for the separation of binary azeotropes by adding an entrainer to alter the properties of the original mixture and then separating the ternary systems. Multi-component azeotropes are refluxed into the column, and the amount and energy consumption of azeotropes are higher for homogeneous azeotropic distillation. The heterogeneous azeotropic distillation has been widely used in industrial production with the advantages of easy separation of the entrainer and the target product, simple operation, low energy consumption, low cost and strong versatility. Liu Wei et al. [9] used heterogeneous azeotropic distillation technology to separate furfural-water system. The results showed that the concentration of furfural products obtained by heterogeneous azeotropic distillation was higher, and the equipment investment and energy consumption cost were reduced compared with conventional furfural-water distillation process. [1]

1.3 Ethanol:

Ethanol is a versatile chemical compound with a wide range of industrial, commercial, and environmental applications. It is most commonly used as a biofuel, either in pure form or blended with gasoline, due to its high-octane number and cleaner combustion properties, which help reduce greenhouse gas emissions. In the chemical industry, ethanol serves as a key solvent and intermediate in the production of various chemicals, pharmaceuticals, and personal care products. It is also used extensively in the manufacture of alcoholic beverages and as a disinfectant in medical and household products [2]. Its renewability and relatively low environmental impact have made ethanol an increasingly important component in the transition toward more sustainable energy and chemical production system [3].

1.4 Objectives:

- -To simulate the azeotropic mixture of water and ethanol (13% Water,87% Ethanol)
- -Using ASPEN HYSYS to compare the two methods of azeotropic distillation.

2.LITERATURE REVIEW

2.1 Distillation process:

The distillation process is carried out by feeding a mixture of liquids to the column at a certain location, called the feed point. The boiler is used to heat the mixture, the boiler is located at the bottom of the column, to form vapor. The liquid flows down through the column, while the vapor rises up. The vapor is cooled in the condenser, located at the top of the column, to form a liquid. The condensed liquid, called the distillate.

2.2 Azeotropic mixture:

An azeotropic mixture is a liquid mixture of definite composition that has the same boiling temperature. When the vapor of the azeotropic mixture is condensed, the resulting liquid mixture has the same composition. In other words, the composition of this mixture cannot be changed by simple distillation. the azeotropic mixture form when the vapor-liquid equilibrium curve (VLE) crosses the 45-degree diagonal line on a phase diagram (fig1,2). The composition of the vapor phase is the same as the composition of the liquid phase at the azeotropic point,

This arises as a result of molecular interactions between different chemical components, in cases where the components are of similar structure and elementary molecules, molecular interactions are minimal, preventing the formation of an azeotrope. for example, hydrocarbon mixtures like (propane, butane, and pentane) do not form any azeotropic behavior because they belong to the same simple phase equilibrium system and do not deviate from Raoult's law to any significant degree.

However, if there is significant dissimilarity between the components, then they may exhibit attractive or repulsive forces toward each other at a certain temperature. Repulsion among the molecules causes an increase in vapor pressure and formation of minimum boiling azeotropes. Activity coefficients in these systems are greater than one indicating positive deviation from Raoult's law. If the repulsion is strong enough, then the mixture could form a later heterogeneous

minimum-boiling azeotrope, which is the term given when two liquid phases co-exist. An important example is the n-butanol-water system, where the repulsion owing to the hydrocarbon chain of butanol from that of the water molecules is strong enough to form two immiscible layers. In other systems, in such instances, the azeotropic mixture boils at temperatures lower than that of the lighter component of the mixture.

Some non-ideality in these systems can be attributed to such a molecular structure. An example of that is from the methanol-water system, where the (-OH) end of methanol molecules have comparable interaction energies with the water molecules leading to a small-scale non-ideality and subsequently slight changes of relative volatility with changing liquid composition, not enough to cause strong azeotropic behavior

However, when the hydrocarbon chain is extended, as in the ethanol-water system, the interaction between the ethyl (-CH₃CH₂) end of the ethanol with water is different from interaction at the hydroxyl end, as the hydrocarbon chain growth occurs in the ethanol-water system. Such interaction causes greater repulsion, resulting in a homogeneous minimum boiling azeotrope that is when a vapor phase is in equilibrium with only one liquid phase.

In contrast, the intermolecular attractive forces between the components reduce the effective vapor pressure of the mixture, resulting in the formation of high-boiling azeotropes. These mixtures exhibit negative deviations from Raoult's law and boil at a higher temperature than any of their pure components. Examples include nitric acid and water, chloroform and acetone, and water and formic acid, where strong intermolecular attractive forces reduce the overall volatility of the mixture.[4]

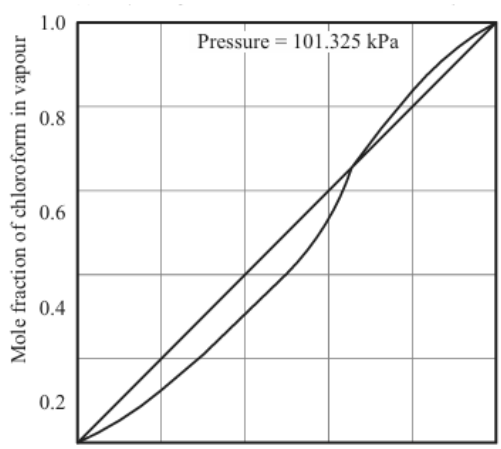


Figure 2. 1: x-y Diagram for Chloroform-Acetone.

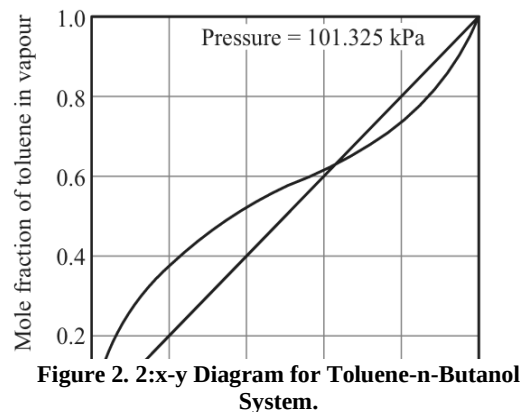


Figure 2. 2:x-y Diagram for Toluene-n-Butanol System.

2.3 Types of azeotropes:

Azeotropes are mixtures that disobey Raoult's law in that their liquid and vapor phases maintain constant composition, rendering ordinary distillation ineffective for separation. They can be characterized based on either their boiling or phase behavior as follows:

2.3.1 Based on Boiling Behaviour:

A-Minimum-boiling azeotropes (Positive azeotrope):

Minimum-boiling azeotropes take place when the total vapor pressure of the mixture exceeds that of any of its components, which creates a boiling point that is lower than what would be expected. This occurs because of low intermolecular forces that increase the volatility of the mix. An example of this is the isopropyl ether-isopropyl alcohol system, where at 1 atm, the azeotropic mixture contains 78 mol% ether and boils at 66°C whereas pure isopropyl ether and isopropyl alcohol boils at 69°C and 82.3°C, respectively. Among known azeotropes, around 90% are of the minimum boiling type

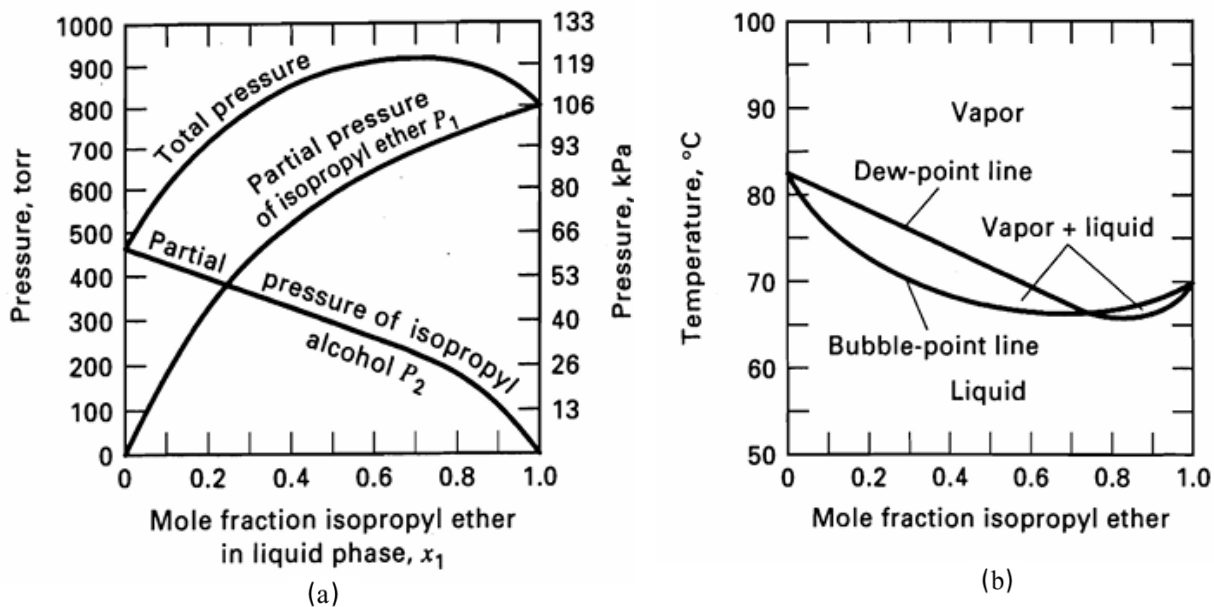


Figure 2. 3: Minimum-boiling azeotropes (Positive azeotrope).

-Minimum-boiling-point azeotrope, isopropyl ether–isopropyl alcohol system:

- (a) partial and total pressures at 70°C;
- (b) vapor–liquid equilibria at 101 kPa;
- (c) phase diagram at 101 kPa.

B-Maximum-boiling azeotropes (Negative azeotrope):

maximum-boiling azeotropes take place when the total vapor pressure of the mixture is lower than any of its components, which creates a boiling point that is higher than either pure component. This behavior emerges from the vigorous intermolecular forces, such as hydrogen bonding or dipole-dipole interactions. These forces reduce volatility and result in a greater boiling point. In distillation, these azeotropes tend to concentrate in the bottom's product rather than the overhead. A well-known example is the chloroform-acetone system, which forms an azeotrope at 64.5°C, even though chloroform and acetone boil at 61.2°C and 56.4°C, respectively. [5]

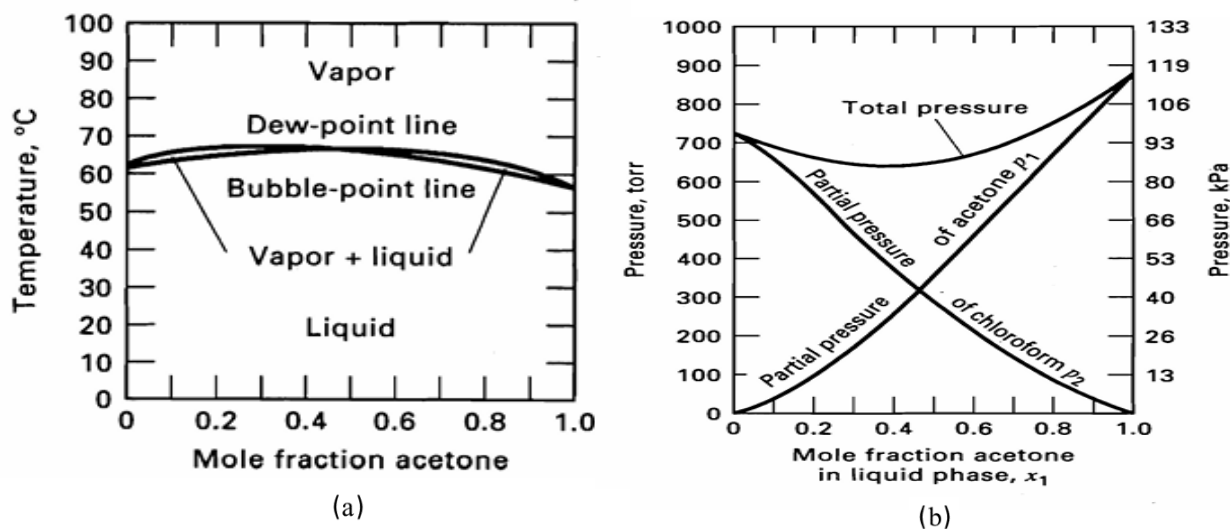


Figure 2. 4: Maximum-boiling azeotropes (Negative azeotrope).

-Maximum-boiling-point azeotrope, acetone–chloroform system:

- (a) partial and total pressures at 60°C.
- (b) vapor–liquid equilibria at 101 kPa.
- (c) phase diagram at 101 kPa pressure.

2.3.2 Based on Phase Behaviour:

A-Homogeneous Azeotropes:

Homogeneous azeotropes exist as a single liquid phase throughout the system. These can be either minimum-boiling, like Ethanol-water system at 101.325 kPa and 78.1°C having 95.6% ethanol (by mass), or maximum-boiling, like the acetone-chloroform system. These azeotropes do not undergo phase separation upon condensation and behave as a single uniform mixture

B-Heterogeneous Azeotropes:

Heterogeneous azeotropes exist as a two liquid phases because of the component's immiscibility. This happens when the activity coefficients are well over 1, and one component separates into a distinct phase, the Heterogeneous azeotropes always demonstrate minimum-boiling behavior, an example is the water-n butanol system, which has a two-phase region (a-b region in Figure 4.8) where total and partial pressures are kept constant but the proportion of the two phases changes. The y-x diagram of this system shows a horizontal line above the immiscible region, demonstrating constant vapor composition, while the phase diagram indicates a minimum constant boiling temperature. This behavior makes the separation easier by decantation prior to distillation.

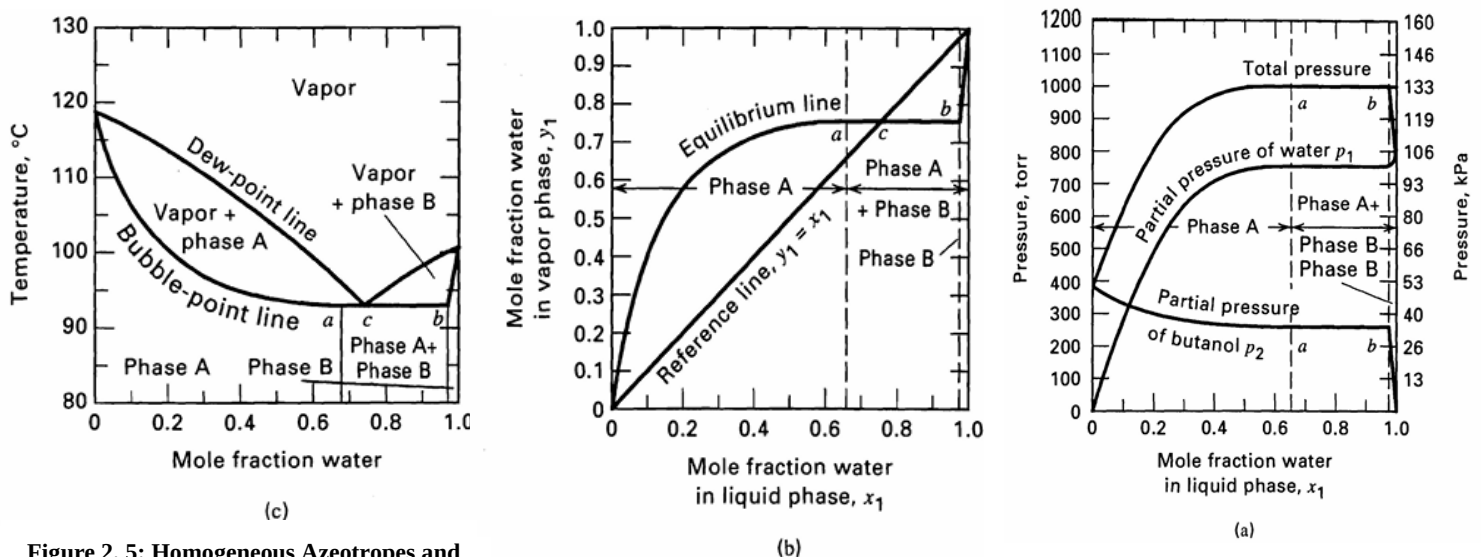


Figure 2. 5: Homogeneous Azeotropes and Heterogeneous Azeotropes

-Minimum-boiling-point (two liquid phases) water/*n*-butanol system:

- (a) partial and total pressures at 100°C
- (b) vapor–liquid equilibria at 101 kPa;
- (c) phase diagram at 101 kPa pressure.

2.4 Azeotropic mixtures separation methods:

Because in the case of an azeotropic mixture the conventional distillation becomes inefficient once the mixture concentration reaches the azeotropic point. Therefore, other methods were used to separate azeotropic mixtures. In general, following are some of the methods, used for separating azeotropic mixtures:

1- Extractive Distillation:

This method involves using a significant quantity of a high-boiling solvent to alter the activity coefficients in the liquid phase of the mixture. This modification enhances the relative volatility of the key components. The solvent is introduced into the column above the feed point, typically a few trays from the top, and exits at the bottom without forming an azeotrope. If the feed mixture is an azeotrope, the solvent disrupts it and can even reverse the volatility of certain components

2. Salt Distillation:

A variation of extractive distillation, this method uses a soluble ionic salt dissolved in the upper reflux to alter the relative volatility of the main components. Since the salt is non-volatile, it remains in the liquid phase throughout the column

3. Pressure-Swing Distillation:

This technique is employed to separate pressure-sensitive azeotropes. It involves two columns operating in sequence at different pressures to achieve the desired separation.

4-Azeotropic Distillation:

a-Homogeneous Azeotropic Distillation:

This method relies on adding an entrainer, which forms a homogeneous azeotrope with one or more components of the mixture. The entrainer is introduced at various points in the column—near the top, with the feed, or near the bottom—depending on where the azeotrope exits

b- Heterogeneous Azeotropic Distillation:

A more efficient method, this process creates a low-boiling, heterogeneous azeotrope using an entrainer. After condensation, the azeotrope separates into two liquid phases. One phase is returned to the column as reflux, while the other is either sent for further separation or used as the final product.

5. Reactive Distillation:

In this method, a separation agent is introduced that selectively and reversibly reacts with one or more components of the mixture. The reaction

product is then separated via distillation, and the reaction is later reversed to recover the original component.

this project focuses on the economic comparison between (pressure swing distillation) and (extractive distillation) of an azeotropic mixture (ethanol - water)

3. PROCESS AND METHODS

There are numerous methods, which are used prevalently in azeotropic distillation for the recovery of azeotropic mixtures. Each method has benefits and is chosen based on the specific characteristics of the treated mixture, hereunder are some widely used methods of which we used two of them.

3.1 PRESSURE SWING METHOD:

Pressure-Swing Distillation (PSD) is an effective method used for the separation of binary azeotropic mixtures. This technique employs two distillation columns that operate at different pressures: the High-Pressure Column (HPC) and the Low-Pressure Column (LPC). The fundamental principle behind PSD is that the relative volatility of components in an azeotropic mixture varies with pressure. By exploiting this pressure dependence, PSD enables separation without the need for an additional entraining agent, which distinguishes it from other methods such as extractive and azeotropic distillation. In PSD, the separation behavior is influenced by the type of azeotrope involved. For maximum-boiling azeotropes, the products of interest are recovered from the distillate streams of the columns, whereas for minimum-boiling azeotropes, the desired products are collected from the bottom streams. A significant advantage of PSD is the possibility of heat integration. Due to the difference in operating pressures, the temperature profiles of the two columns are distinct. This allows the overhead vapor from the HPC to be used as a heat source for the reboiler of the LPC, enhancing the energy efficiency of the process.[6][7]

The typical steps involved in the PSD process are as follows:

1. **Feed Introduction:** The binary azeotropic mixture is introduced into one of the distillation columns, typically the Low-Pressure Column (LPC).
2. **Initial Separation at Low Pressure:** The mixture undergoes partial separation under low pressure, based on the altered relative volatilities.
3. **Secondary Separation at High Pressure:** The partially separated stream is transferred to the High-Pressure Column (HPC), where further separation occurs due to the different vapor-liquid equilibrium at higher pressure.
4. **Product Recovery:** The target components are recovered either from the distillate (for maximum-boiling azeotropes) or from the bottom stream (for minimum-boiling azeotropes), depending on the nature of the azeotrope.
5. **Recycle Stream Management:** The non-product stream may be recycled back into the process to improve overall separation efficiency.

6. **Heat Integration:** The temperature difference between the columns enables the overhead vapor from the HPC to serve as a heat source for the LPC reboiler, reducing external energy demand.

This method is particularly advantageous in industrial applications where avoiding the use of additional separation agents and minimizing energy consumption are critical considerations.

3.2 Entrainer Method:

The entrainer method is a widely used technique in the chemical industry for the separation of azeotropic mixtures—mixtures of two or more liquids that form a constant boiling point composition, making them difficult to separate by simple distillation. This method involves the intentional addition of a third component, known as an **entrainer** or **azeotrope breaker**, which alters the vapor-liquid equilibrium of the system. The entrainer is selected such that it preferentially forms a new azeotrope with one of the original components, typically resulting in the formation of a heterogeneous azeotrope with a lower boiling point than the original mixture. The resulting azeotrope can then be separated by conventional distillation methods.

A common example of this technique is the addition of cyclohexane to an ethanol-water azeotropic mixture. Cyclohexane forms a low-boiling ternary azeotrope with ethanol and water, which separates as a vapor and can be condensed and collected, thereby allowing the recovery of nearly pure ethanol from the remaining mixture. The entrainer method is particularly advantageous in cases where azeotropic distillation or pressure-swing distillation are less effective or when specific product purity is required. It is also adaptable to both batch and continuous operations.[8]

The general steps involved in the entrainer method are as follows:

1. **Entrainer Addition:** The azeotropic mixture is placed into a distillation system, and a suitable entrainer is added. The entrainer is chosen based on its ability to form a new azeotrope with one of the mixture's components.
2. **Heating and Vaporization:** The combined mixture is heated to its boiling point. As the mixture boils, the entrainer interacts with one of the original components to form a new azeotrope.

3. **Fractionation:** The vaporized mixture passes through a fractionating column. Due to differences in boiling points and azeotrope formation, the newly formed azeotrope is separated from the remaining component(s).
4. **Condensation and Separation:** The azeotropic vapor is condensed and collected separately, often as a heterogeneous mixture that can be further separated by decantation.
5. **Product Recovery:** The remaining component in the distillation flask, now largely free of the azeotrope-forming component, is collected. If needed, it can undergo additional purification steps to achieve the desired purity.
6. **Entrainer Recovery (Optional):** In many industrial systems, the entrainer is recovered from the azeotrope mixture and recycled back into the process, enhancing economic and environmental sustainability.

This method is commonly employed in the production of high-purity solvents, fuels, and specialty chemicals, where azeotropes would otherwise limit product quality. Its flexibility and effectiveness make it a vital technique in both research and large-scale chemical processing.[9]

Possible entrainers

There are many possible entrainers that can be used: Benzene, pentane, hexane, acetone, heptane, dimethyl ether and cyclohexane (which we used in this project).

3.3 Cyclohexane:

Cyclohexane is widely employed as an entrainer in azeotropic distillation due to its favorable physicochemical properties. It readily forms heterogeneous azeotropes with a variety of organic compounds, thereby enabling the disruption of azeotropic behavior and facilitating the separation of individual components through conventional distillation. Its high miscibility with numerous organic solvents broadens its applicability across diverse mixture systems. Additionally, cyclohexane possesses a boiling point that closely aligns with many target compounds, enhancing its compatibility in separation processes. From an environmental and safety perspective, cyclohexane is relatively non-toxic and chemically inert, reducing potential hazards during handling and operation. Furthermore, its widespread availability and low cost contribute to its practicality and cost-effectiveness in both laboratory-scale and industrial azeotropic separation processes.[10]

4. SIMULATION AND SOFTWARE WORK

4.1 PROCESS SIMULATION:

Process simulation plays a critical role in modern chemical engineering by enabling the analysis, design, and optimization of chemical processes before physical implementation. Through the use of computational models and simulation software, engineers can predict the behavior of complex systems under various operating conditions, identify optimal process parameters, and evaluate performance metrics such as yield, energy consumption, and cost. This virtual prototyping reduces the need for extensive experimental trials, saving both time and resources while enhancing safety and sustainability. Additionally, process simulation facilitates the integration of unit operations, the assessment of control strategies, and the troubleshooting of existing systems. In both research and industrial contexts, it serves as an indispensable tool for decision-making, scale-up, and continuous improvement of chemical processes.

4.2 ASPEN HYSYS:

Aspen HYSYS is a process simulation tool that leads in chemical engineering for simulating steady-state and dynamic processes. It permits engineers to design, optimize, and analyze various operations such as distillation, heat exchange, and chemical reactions. Aspen HYSYS features a comprehensive thermodynamic database and is user-friendly with an intuitive interface, making it effective in creating processes, incorporating energy, and troubleshooting and therefore used both for research in academia and industry.

4.3 PRESSURE SWING METHOD USING ASPEN HYSYS:

The azeotrope in the ethanol-water system presents a barrier to separation, but pressure swing distillation can be used to purify ethanol. A technically feasible design for purifying ethanol to 99mol-% with pressure swing distillation can be constructed using Aspen HYSYS.

Problem Description:

This work aims at obtaining high purity ethanol for a binary feed of ethanol and water by simulating the pressure swing distillation column.

- **Feed:** 100 kgmole/h with 87 mol-% ethanol, 13 mol-% water.
- **LPC:** operating pressure 1.2 bar; no. of stages = 30.
- **HPC:** operating pressure 20 bar; no. of stages = 30.[11]

Simulation [12]:

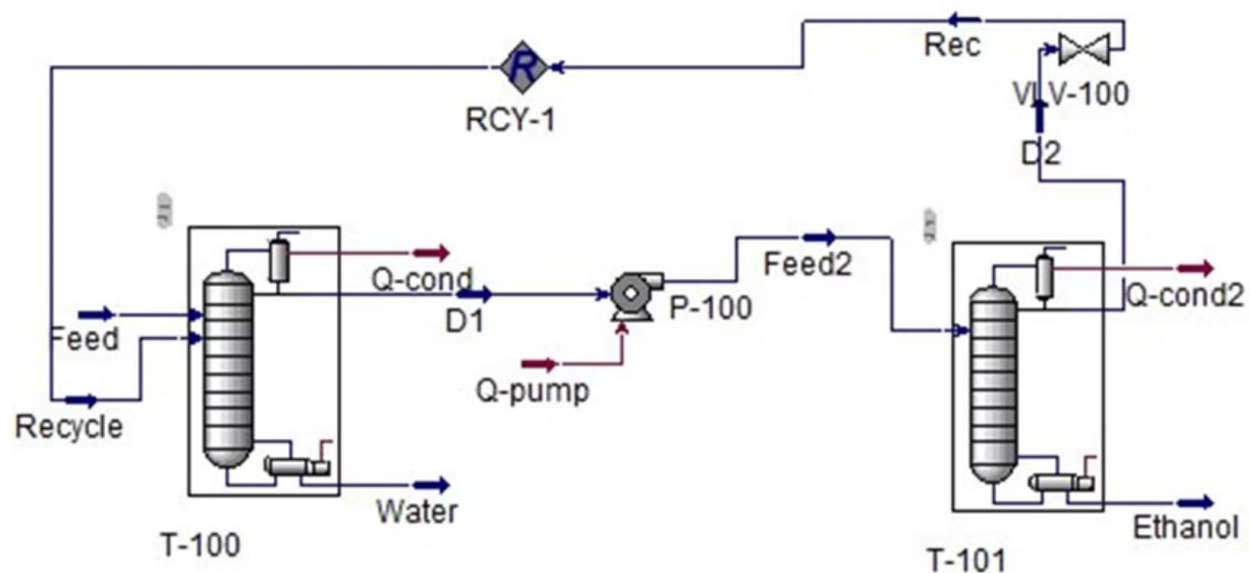


Figure 4. 1: Pressure Swing Method

LPC		
Number Of Stages	30	
Feed Stage	25	
Recycle Inlet Stage	25	
Pressure	120	KPa

Feed		
Temperature	65	°C
Pressure	120	KPa
Molar Flow	100	Kmole/H

HPC		
Number Of Stages	30	
Feed Stage	25	
Pressure	2000	KPa

Ethanol		
Temperature	179.7	°C
Pressure	2000	KPa
Molar Flow	53	Kmole/H

4.4 ENTRAINER METHOD USING ASPEN HYSYS:

In this example, cyclohexane is used as the entrainer to separate water and ethanol to produce anhydrous ethanol. By using the proper amount of solvent, we obtain pure ethanol from the bottom of the first column. The stream from the top of the first column is separated into two streams using a three-phase condenser: One stream is solvent rich and is recycled back to the first column as solvent; the other stream is well within another distillation region so that we can use the second column to obtain pure water. The top stream of the second column is recycled back to the first column as feed.

Problem Description:

Design a separation train to achieve ethanol purity of >99.95 mol-% and water effluent stream with 99 mol-% purity.

- **Feed:** 100 kgmole/h with 87 mol-% ethanol, 13 mol-% water.
- **Addition:** Cyclohexane is added to the first column.
- **Separation:** Ethanol and water streams are separated through distillation.
- **Recycle:** Recycle streams are managed between the two columns.

Simulation:

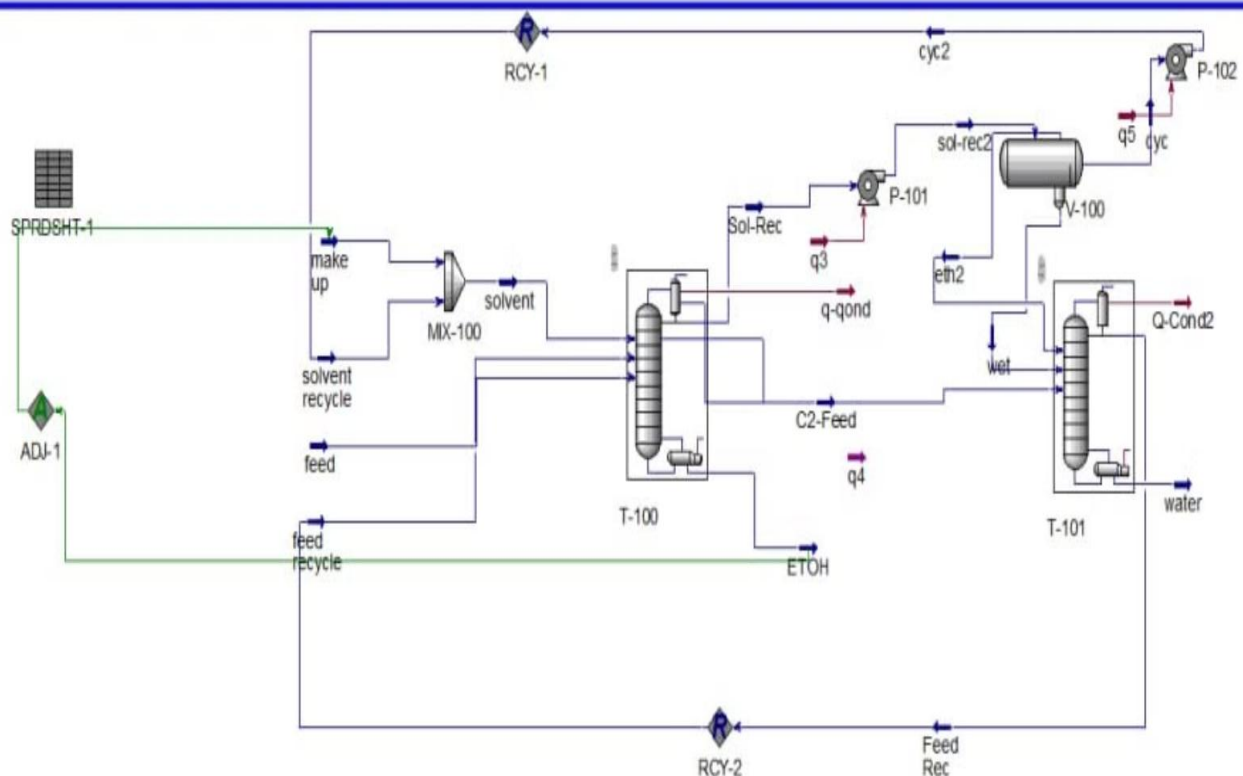


Figure 4. 2 : Entrainer Method.

T-100		
Number of Stages	62	
Feed Stage	20	
Recycle Inlet Stage	20	
Solvent Inlet stage	1	
Pressure	100	KPa

T-101		
Number of Stages	50	
Feed Stage	35	
Pressure	100	KPa

Ethanol		
Temperature	77.92	°C
Pressure	100	KPa
Molar Flow	86.86	Kmole/H

Feed		
Temperature	78.45	°C
Pressure	100	KPa
Molar Flow	100	Kmole/H

5. ECONOMIC ANALYSIS AND RESULTS

5.1 Results:

This section presents a comparison between Pressure Swing Distillation (PSD) and Extractive Distillation (ED) for separating an ethanol-water azeotropic mixture.

Both systems had feed streams with equivalent composition of (20% water by mole, 100 kmol/h total feed).

The final ethanol product concentrations and flow rates was calculated as:

- ED: 86 kmol/h rate of product and a concentration of 99.9% pure ethanol.
- PSD: 53 kmol/h rate of product and a concentration of 99% pure ethanol.

The evaluation considers capital and operational costs, energy consumption, environmental impact, and product purity.

5.2 Economic Analysis:

1-Cost Analysis:

Table 1: Cost Analysis

Factor	PSD method	ED method	
capital cost	6,583,640	6,346,070	USD
operating cost	3,201,570	4,966,260	USD/year
utility cost	1,518,030	3,256,480	USD/year
total installed cost	3,196,800	2,674,300	USD
Entrainer cost	0	27,007	USD/year
Total Cost	14,500,040	17,270,117	USD

The capital cost for both methods is nearly the same, although slightly higher in the (PSD)method The operating cost for the PSD is lower than that of ED, mainly due to the absence of entrainer purchase and handling. However, ED has a significantly higher utility cost, indicating more energy input (especially due

to the solvent recovery system), whereas PSD in this case appears more utility-efficient. ED incurs a minor but non-zero annual entrainer cost, while PSD does not require any additional chemicals.

2-Energy Consumption and Environmental Impact:

Table 2: Energy Consumption and Environmental Impact

factor	PSD method	ED method	
energy consumption	18.9	22	MW
Carbon Emissions	3987	4379	Kg/hr
recycling effectiveness	moderate	High	-----

Despite the ED method consuming more energy (22 MW vs. 18.9 MW), it also shows higher recycling efficiency leading to more effective reuse of the entrainer and possibly better long-term sustainability. However, the carbon emissions of the ED method are higher than PSD method indicating a greater environmental footprint unless renewable energy sources are used.

3-Product Quality and Production Rate:

Table 3: Product Quality and Production Rate

factor	PSD method	ED method	
final ethanol purity	99%	99.95%	-----
product rate (kmol/hr)	53	86	Kmol /hr

The product yield of ED (86 kmol/hr) is significantly higher than that of PSD (53 kmol/hr), making ED method more productive for the same feed input. In terms of purity ED method also achieves 99.9% pure ethanol slightly superior to PSD method 99% pure ethanol. Therefore, despite the energy demands, ED method delivers a higher quality and product rate which is beneficial for fuel ethanol, pharmaceuticals, and high-grade industrial applications. The ED method also benefits more from recycling (cyclohexane recovery), whereas PSD method recycling stream increases energy demand for higher purity.

6. DISCUSSION AND CONCLUSION

6.1 Discussion and Conclusion

When comparing the two methods, the Pressure Swing Distillation method (PSD) outperformed Extractive Distillation method (ED) across economic and environmental impact, while the ED had better product quality.

ED achieved higher product yield (86 kmol/h vs. 53 kmol/h) with superior purity (99.9% vs. 99%). It operated at higher energy consumption and emitted more of the carbon dioxide. The presence of an entrainer (cyclohexane) introduced an additional cost of (\$27,007/year), but this was offset by the recycle system

On the other hand, the PSD method, in addition to the simplicity of not requiring an entrainer, incurred lower energy consumption to operate two columns at different pressures and handling large amounts of vapor. with lower utility and operating cost its environmental footprint remained less than ED method.

From a capital cost perspective, the difference was very small. However, total cost of ownership (including utilities and operation) favors the PSD method however, the ED system proved more adaptable for heat integration and had fewer scaling limitations due to its lower temperature and pressure requirements, moreover it produce higher collate and product rate

In conclusion, Pressure Swing Distillation method (PSD) presents itself as the more sustainable, cost-effective option for ethanol-water azeotropic separation under the studied conditions. while the Extractive Distillation method (ED) is more efficient and more suitable for high purity ethanol production required in pharmaceutical and fuel-grade applications.

References:

- [1] Roman, C., & García-Morales, M. (2019). Achieving a better understanding of binary azeotropic mixtures distillation through Aspen Plus process simulations. **Computer Applications in Engineering Education**, **27**(6), 1453–1464. <https://doi.org/10.1002/cae.22161>
- [2] Sun, J., & Wang, Y. (2014). Recent Advances in Catalytic Conversion of Ethanol to Chemicals. **ACS Catalysis**, **4**(4), 1078–1090. <https://doi.org/10.1021/cs4011343>
- [3] Dagle, R. A., et al. (2020). Ethanol as a Renewable Building Block for Fuels and Chemicals. **Industrial & Engineering Chemistry Research**, **59**(11), 4843–4853. <https://doi.org/10.1021/acs.iecr.9b05729>
- [4] Seader, J. D., Henley, E. J., & Roper, D. K. (2011). **Separation process principles: Chemical and biochemical operations** (3rd ed.). John Wiley & Sons.
- [5] Thakore, S. B., & Bhatt, B. I. (2007). **Introduction to Process Engineering and Design** (2nd ed.). McGraw-Hill Education.
- [6] Luyben, W. L. (2014). Methanol/Trimethoxysilane azeotrope separation using pressure-swing distillation. **Industrial & Engineering Chemistry Research**, **53**, 5590–5597.
- [7] Wang, Y., et al. (2015). Control of Heat Integrated Pressure-Swing-Distillation Process for Separating Azeotropic Mixture of Tetrahydrofuran and Methanol. **Industrial & Engineering Chemistry Research**, **54**(5), 1646–1655. <https://doi.org/10.1021/ie505024q>
- [8] Laroche, L., et al. (1991). Homogeneous azeotropic distillation: Comparing entrainers. **The Canadian Journal of Chemical Engineering**, **69**(6), 1302–1319. <https://doi.org/10.1002/cjce.5450690611>
- [9] Laroche, L., et al. (1991). Homogeneous azeotropic distillation: Comparing entrainers. **The Canadian Journal of Chemical Engineering**, **69**(6), 1302–1319. <https://doi.org/10.1002/cjce.5450690611>
- [10] Chien, I., Zeng, K., & Chao, H. (2004). Design and Control of a Complete Heterogeneous Azeotropic Distillation Column System. **Industrial & Engineering Chemistry Research**, **43**(9), 2160–2174. <https://doi.org/10.1021/ie030406v>
- [11] Aspen Technology, Inc. (2012). **Aspen HYSYS user guide**. AspenTech.
- [12] Aspen Technology, Inc. (Year). **Aspen HYSYS (Version X.X) [Computer software]**. Retrieved from.

List of Figures:

Figure 2. 1: x-y Diagram for Chloroform-Acetone.	11
Figure 2. 2:x-y Diagram for Toluene-n-Butanol System.	11
Figure 2. 3: Minimum-boiling azeotropes (Positive azeotrope).	12
Figure 2. 4: Maximum-boiling azeotropes (Negative azeotrope).	13
Figure 2. 5: Homogeneous Azeotropes and Heterogeneous Azeotropes	14
Figure 4. 1: Pressure Swing Method	23
Figure 4. 2 : Entrainer Method.	24

List of Tables:

Table 1: Cost Analysis	27
Table 2: Energy Consumption and Environmental Impact	28
Table 3: Product Quality and Production Rate	28